

Problem C. Endless Holidays

Time Limit 3000 ms

Problem Statement

The Kingdom of AtCoder has N cities, called city $1, 2, \dots, N$. There are M bidirectional roads connecting pairs of cities, where the i -th road connects cities U_i and V_i . Any pair of cities can be reached from each other by traversing some roads.

In the Kingdom of AtCoder, a week consists of W days. A week proceeds through days $1, 2, \dots, W$, and the day after day W is day 1.

Each city has certain days of the week that are holidays. The holiday information for city i is given as a string S_i of length W : if the j -th character of S_i is `o`, day j is a holiday; if it is `x`, day j is a weekday.

Takahashi chooses a city he likes and visits it at noon on day 1. Each night thereafter, he repeatedly chooses to either stay in his current city or move to a city directly connected by a road. Output `Yes` if it is possible for him to keep moving so that the city he is in at noon every day is a holiday, and `No` otherwise.

T test cases are given; solve each of them.

Constraints

- $1 \leq T \leq 10^5$
- $1 \leq N \leq 10^5$
- $N - 1 \leq M \leq 10^5$
- $1 \leq U_i < V_i \leq N$
- Any pair of cities can be reached from each other by traversing some roads.
- $2 \leq W \leq 10$
- T, N, M, U_i, V_i, W are integers.
- S_i is a string of length W consisting of `o`, `x`.
- The sum of N over all test cases is at most 10^5 .
- The sum of M over all test cases is at most 10^5 .

Input

The input is given from Standard Input in the following format:

```
 $T$   
case1  
case2  
⋮  
case $T$ 
```

Here, case _{i} denotes the input for the i -th test case. Each test case is given in the following format:

```
 $N$   $M$   
 $U_1$   $V_1$   
 $U_2$   $V_2$   
⋮  
 $U_M$   $V_M$   
 $W$   
 $S_1$   
 $S_2$   
⋮  
 $S_N$ 
```

Output

Output T lines. The i -th line should contain the answer for the i -th test case.

Sample 1

Input	Output
3 4 4 1 2 1 4 2 4 2 3 3 xx0 x0x 0x0 0xx 1 0 4 0000 5 5 1 4 2 3 4 5 3 4 2 5 7 0xxxxxx xx0xxx0 xxx0x0x x0xx0xx 0xxx0xx	Yes Yes No

For the first test case, for example, the condition can be satisfied by repeatedly moving along cities $4 \rightarrow 2 \rightarrow 1 \rightarrow 4 \rightarrow 2 \rightarrow 1 \rightarrow \dots$. Alternatively, the condition can also be satisfied by repeatedly moving along cities $3 \rightarrow 2 \rightarrow 3 \rightarrow 3 \rightarrow 2 \rightarrow 3 \rightarrow \dots$.

For the second test case, the condition can be satisfied by staying in city 1 indefinitely.

For the third test case, it is impossible to keep moving while satisfying the condition.